Securitization and Leverage in General Equilibrium

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Motivation

- Securitization (pooling of debt contracts) widely used in reality and played crucial role in financial crisis of 2007–2009.
- But few (applied) GE models have collateralized lending, default, and asset-backed securities as key elements.
- Without a clear theoretical and quantitative benchmark, cannot ask or answer normative questions.

Questions

1. Can we share idiosyncratic risk without requiring a lot of information?
2. What are effects of collateralized lending, default, and asset-backed securities on growth, welfare, and inequality?
3. What contracts are traded in equilibrium?

Contribution

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   - in equilibrium only lowest collateral (highest leverage) loans are traded $\implies$ endogenous emergence of "subprime loans".

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   - in equilibrium only lowest collateral (highest leverage) loans are traded $\implies$ endogenous emergence of "subprime loans".
3. (Numerical) As collateral requirement loosens (more risk sharing), borrowing rate $\uparrow$, saving $\downarrow$, welfare $\uparrow$ if no moral hazard. Effect on growth ambiguous.

Literature

Positive role of default Dubey, Geanakoplos, & Shubik (2005), Zame (1993).

Collateral equilibrium Geanakoplos & Zame (2014), Geanakoplos (1997), Araújo et al (2000, 2002, 2005), Kubler & Schmedders (2003), Steinert & Torres-Martínez (2007), Cao (2010).

Tractable GEI with heterogeneous agents Saito (1998), Calvet (2001), Krebs (2003a, 2003b, 2006), Angeletos & Calvet (2005, 2006), Angeletos (2007), Toda (2014).

How is this paper different?

Mathematical Economics (GEI) Dubey, Geanakoplos, & Shubik (2005), Zame (1993), Geanakoplos & Zame (1995), Bisin & Gottardi (1999), Citanna & Villanacci (2004), etc.
$\implies$ Complex financial structure but analytically intractable (few quantitative applications).

DSGE with Heterogeneous Agents Krusell & Smith (1998), Kubler & Schmedders (2003), Chatterjee et.al. (2007), Cao (2010), etc.
$\implies$ Quantitative but simple financial structure (due to 'curse of dimensionality').

This Paper Complex financial structure but still highly tractable and quantitative!

Outline

1. Introduction to ABS
2. Theoretical analysis
3. Quantitative analysis

Mechanics of securitization

1. Bank originating loans sells individual loans to a trust called *special purpose vehicle* (SPV).
2. SPV pools debt contracts and creates tranches of claims to the pool according to seniority to be repaid (securitization & tranching).
3. SPV sells ABS to investors (usually keeps most junior tranche to avoid moral hazard issue); bank keeps servicing loans and pass through proceeds to SPV after deducting service fees.

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- In this paper, treat bank and SPV as one entity (financial intermediary).

Three roles of securitization

1. Diversification of idiosyncratic risks: by pooling loans with similar characteristics, only aggregate risk remains.
2. Lowering informational costs and increasing liquidity: due to pooling, buyer of ABS suffices to know the average characteristics of loans. (Though may lead to adverse selection.)
3. Credit enhancing: by splitting cash flow of pool into junior and senior tranches, can create relatively safe asset.

Agents

- Time: $t = 0, 1$.
- Continuum of agents with mass 1, indexed by $i \in I = [0, 1]$.
- Preference is identical homothetic CRRA with disutility from effort $e \in E$:

$$U(c_0, e, c_1) = f\left(c_0, e, \mathrm{E}[c_1^{1-\gamma}]^{\frac{1}{1-\gamma}}\right),$$

  where $f(c, e, v)$: homogeneous of degree 1 in (c, v).
- Endowment: agent i endowed wealth (capital) $w_i > 0$ at $t = 0$, nothing thereafter.
- Financial intermediary services loans and asset-backed securities (ABS) (coming shortly).

Technology

- Technology: AK model with agent-specific productivities $(A_i)_{i \in I}$, *e.g.*, private equity, human capital, growing potatoes in backyard, etc. ▸ relevance

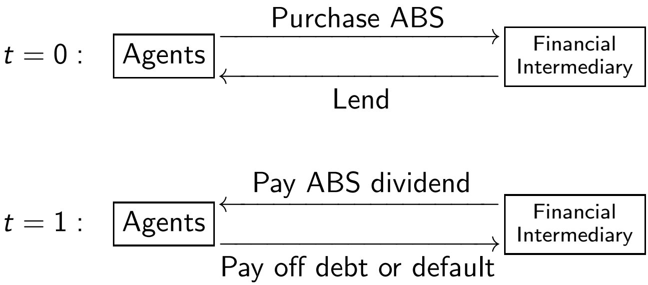
end of time 0	$\rightarrow$	beginning of time 1
invest k_i	$\rightarrow$	get $A_i k_i$ in total

- Productivity A_i is i.i.d. across agents, conditional on effort and aggregate shock. Unknown at $t = 0$, and realizes at $t = 1$.
- Markets are incomplete: agents cannot directly insure against the idiosyncratic component of A_i because private information.

Financial structure

- Perfectly competitive, risk neutral financial intermediaries offer non-recourse loans from a menu $L = \{1, \dots, L\}$ (*i.e.*, no penalty of default other than confiscation of collateral).
- Loan $l \in L$ characterized by gross borrowing rate $R_b^l \geq 0$ (determined in equilibrium) and exogenous collateral requirement $c^l \geq 1$.
- To borrow \$1 from loan l, must invest $\$c^l$ in technology and put up its product $A_i c^l$ as collateral.
- Financial intermediary pools all debt contracts of same type and issues asset-backed securities (ABS) $l = 1, \dots, L$ to raise capital.

Flow of capital



Return on ABS

- Because no penalty of default, agent i delivers $\min\left\{A_i(e_i)c^l, R_b^l\right\}$ for each dollar taken from loan l.
- By perfect competition among intermediaries, ABS return is cross-sectional average of individual deliveries,

$$\begin{aligned} & R^l_{\mathrm{ABS}}(R^l_b; A) \\ & \quad := \mathrm{E}\left[R^l_b 1\left\{A_i c^l \geq R^l_b\right\} + A_i c^l 1\left\{A_i c^l < R^l_b\right\} \,\middle|\, A\right]. \end{aligned}$$

(Implicitly depends on distribution of effort $\{e_i\}$.)
- Agents need not keep track of individual states because ABS returns have no idiosyncratic risk.
- Financial intermediary suffices to verify collateral of defaulters $\implies$ Weak informational requirement.

Budget and portfolio constraints

- $\theta, \phi^l, \psi^l \geq 0$: fraction of wealth invested in technology, ABS l, and borrowed from loan l. $\pi = (\theta, \phi, \psi) \in \mathbb{R}_+^{1+L+L}$: portfolio.
- Gross return on portfolio for agent i:

$$R_i(\pi, e) = A_i(e)\theta + \sum_{l=1}^{L} R_{\mathrm{ABS}}^l \phi^l - \sum_{l=1}^{L} \min\left\{A_i(e)c^l, R_b^l\right\} \psi^l.$$

- Portfolio constraint:

$$\Pi = \left\{ \pi = (\theta, \phi, \psi) \in \mathbb{R}_+^{1+L+L} \;\middle|\; \begin{array}{c} \theta + \sum_{l=1}^{L}(\phi^l - \psi^l) = 1, \\ \theta \geq \sum_{l=1}^{L} c^l \psi^l \end{array} \right\}.$$

- Budget constraint: $c_1 = R_i(\pi, e)(w_i - c_0)$.

Equilirium concept

Two possible equilibrium concepts:

Collateral equilibrium with exogenous financial markets
: Intermediaries passively offer loans, entrepreneurs optimize, and markets clear.

Collateral equilibrium with endogenous financial markets
: Intermediaries choose loans to maximize profits, entrepreneurs optimize, and markets clear.

Collateral equilibrium

Definition (Collateral equilibrium)

Given initial wealth $(w_i)_{i \in I}$ and collateral requirement $(c^l)_{l=1}^L$, the individual consumption, effort, and portfolio choice $(c_{i0}, c_{i1}, e_i, \pi_i)_{i \in I}$ and borrowing rates $(R_b^l)_{l=1}^L \subset [0, \infty]^L$ constitute a *collateral equilibrium* if

1. each agent maximizes utility subject to budget and portfolio constraints, and
2. for each loan type $l \in L$ lending and borrowing are matched:

$$\int_I \phi_i^l (w_i - c_{i0}) \mathrm{d}i = \int_I \psi_i^l (w_i - c_{i0}) \mathrm{d}i,$$

3. (each intermediary maximizes profits.)

Individual decision

- Individual optimal consumption-portfolio problem reduces to

$$V_i(w) := \max_{\substack{0 \le c \le w \\ e \in E \\ \pi \in \Pi}} f\left(c, e, \mathrm{E}[R_i(\pi, e)^{1-\gamma}]^{\frac{1}{1-\gamma}}(w - c)\right).$$

- This problem can be split into

$$\rho(e) := \max_{\pi \in \Pi} \mathrm{E}[R(\pi, e)^{1-\gamma}]^{\frac{1}{1-\gamma}},$$

$$V(w) = \max_{\substack{0 \le c \le w \\ e \in E}} f(c, e, \rho(e)(w - c)).$$

- Since optimal portfolio problem common across all agents, may focus on symmetric equilibrium (common consumption rate, effort, and portfolio choice).

Existence

Theorem (Existence)

Let everything be as above and

$$\Pi' = \left\{ \pi = (\theta, \phi, \psi) \in \mathbb{R}_+^{1+L+L} \,\middle|\, \theta = 1, \phi = \psi, \theta \geq \sum c^l \psi^l \right\}$$

be the portfolio constraint with matched borrowing and lending (clearly nonempty, compact, and convex). Suppose that for all borrowing rates $(R_b^l)_{l=1}^L \in [0, \infty]^L$ and equilibrium effort $\bar{e} \in E$ we have

$$\mathrm{E}\left[\sup_{e \in E, \pi \in \Pi'} R(\pi, e; \bar{e})^{1-\gamma} \right] < \infty.$$

Then a collateral equilibrium exists.

Portfolio property

Theorem (Maximum leverage)

Let everything be as above. Then in equilibrium only the lowest collateral loan is active and agents leverage to maximum. More precisely, the collateral equilibrium portfolio $\pi^* = (\theta^*, \phi^*, \psi^*)$ *satisfies* $\theta^* = 1$ *and*

$$\phi^{*l} = \psi^{*l} = \begin{cases} 1/c^l, & (c^l = \min_{l'} c^{l'}) \\ 0. & (c^l > \min_{l'} c^{l'}) \end{cases}$$

- Intuition: maximum leverage allows to sell off maximum downside idiosyncratic risk (although higher borrowing rate).
- Endogenous emergence of "subprime" loans.

Welfare property

Proposition

Let $\left\{(c_{i0}, c_{i1}, e^*, \pi^*)_{i\in I}, (R_b^l)_{l=1}^L\right\}$ be a symmetric collateral equilibrium, where $\pi^* = (\theta^*, \phi^*, \psi^*)$. Then the utility of every agent is a monotonic function of $\mathrm{E}[R(\pi^*, e^*; e^*)^{1-\gamma}]^{\frac{1}{1-\gamma}}$, hence every agent agrees on policies. Furthermore, the collateral equilibrium allocation $(c_{i0}, c_{i1})_{i\in I}$ weakly Pareto dominates the GEI allocation, and strictly Pareto dominates it only if active default occurs.

- Welfare analysis is unambiguous despite heterogeneous agents.
- If multiple equilibria, all equilibria are Pareto-ranked.
- With no effort, can show equilibrium is "constrained efficient".

Zero net supply assets with aggregate payoffs

- So far ABS are the only assets.
- Assume also assets in zero net supply indexed by $k \in K = \{1, \dots, K\}$.
- Asset k delivers $D^k(A)$ (depends only on aggregate shock) in period 1. E.g., $D^k(A) = 1$: risk-free bond, $D^k(A) = \max\left\{A - S^k, 0\right\}$: call option with strike S^k.
- Portfolio: $(\theta, \phi_{\mathrm{ABS}}, \phi_0, \psi) \in \mathbb{R}_+ \times \mathbb{R}_+^L \times \mathbb{R}^K \times \mathbb{R}_+^L$.
- Can define equilibrium by market clearing for all assets.

Asset pricing and irrelevance of zero net supply assets

Theorem

If $\left\{(c_{i0}, c_{i1}, e_i, \theta_i, \phi_{\mathrm{ABS},i}, \phi_{0i}, \psi_i)_{i\in I}, (R_b^l)_{l=1}^L, (P^k)_{k=1}^K\right\}$ *is a collateral equilibrium with zero net supply assets, there exists a collateral equilibrium with the same allocation, borrowing rates, asset prices, and a common portfolio choice* $\pi^* = (\theta^*, \phi^*_{\mathrm{ABS}}, \phi^*_0, \psi^*)$ *with no trade in the zero net supply assets:* $\phi^*_0 = 0$. *Asset price is*

$$P^k = \frac{\mathrm{E}[R(\pi^*)^{-\gamma} D^k]}{\mathrm{E}[R(\pi^*)^{1-\gamma}]}.$$

Tranching

- In reality, ABS are also created by splitting other ABS into senior and subordinate tranches (prime, mezzanine, junk, etc.).
- Suppose ABS l is split into tranches $t \in \{1, \dots, T_l\}$, with (exogenous) payments $(D^{lt}(A))_{t=1}^{T_l}$ and (endogenous) prices $(P^{lt})_{t=1}^{T_l}$.
- Can define equilibrium as before.

Irrelevance of tranching

Proposition

If $\left\{(c_{i0}, c_{i1}, e_i, \theta_i, \phi_i, \psi_i)_{i \in I}, (R^l_b)^L_{l=1}, ((P^{lt})^{T_l}_{t=1})^L_{l=1}\right\}$ *is a collateral equilibrium with tranching, there exists a collateral equilibrium with no tranching with the same allocation, borrowing rates, and a common portfolio choice* $\pi^* = (\theta^*, \phi^*, \psi^*)$*.*

- Zero net supply assets and tranching are irrelevant for risk sharing.
- In particular, with no collateralized lending ($L = \emptyset$, GEI), equilibrium allocation is autarky.
- ABS attains risk sharing without stringent informational requirement that GEI cannot.

Full model

- Basic model generalizes to
  1. multiple technologies,
  2. multiple periods.
- See paper for full model.
- Now is good time to ask questions.

Quantitative analysis

- Infinite horizon model with Epstein-Zin preferences

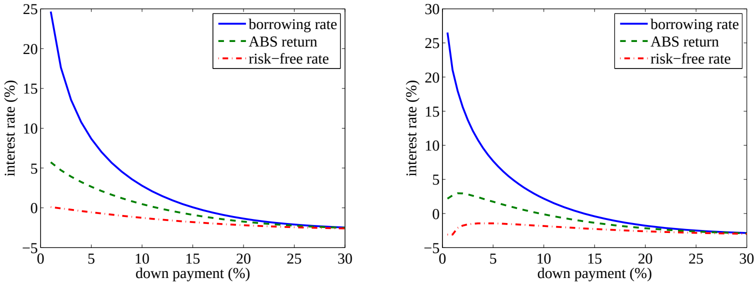
$$f(c,e,v)=((c(1-e)^{\alpha})^{1-\sigma}+\beta v^{1-\sigma})^{\frac{1}{1-\sigma}}.$$

- $\beta=0.96$, RRA $\gamma=3$, EIS $1/\sigma=0.9$, $\alpha=0$ or 0.1.
- 1 or 2 technologies, one risky with productivity

$$\log A_i(e)\sim N(\mu(e)-v^2/2,v^2),$$

  the other risk-free ($r_f=0.01$). Expected returns $\mu(e)=\mu_{\max}e+\mu_{\min}(1-e)$, aggregate & idiosyncratic volatility v_a,v_i with $v^2=v_a^2+v_i^2$.
- $\mu_{\max}=0.07$, $\mu_{\min}=0.03$, $v_a=0.15$, $v_i=0.1$.

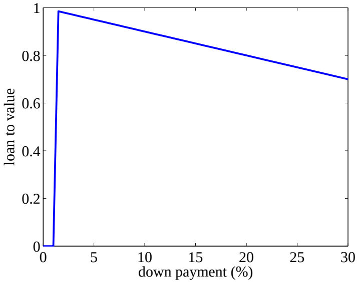
Interest rates and returns



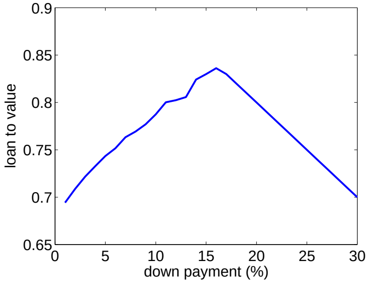
(a) $\alpha = 0$ (no moral hazard) (b) $\alpha = 0.1$ (moral hazard)

- Borrowing rates sharply increase with lower down payments.
- Risk-free rate remains low (-2–0%) but high default premium.

Loan-to-value (ψ^*/θ^*)



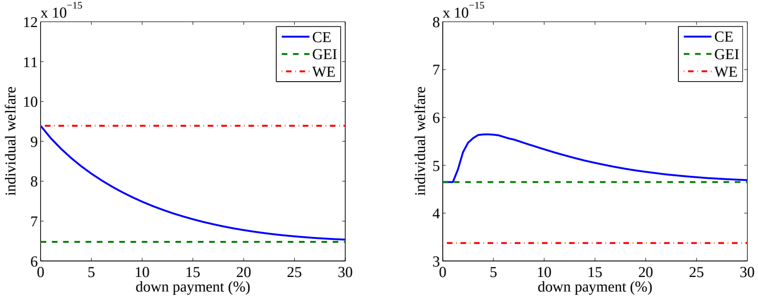
(c) With moral hazard.



(d) With no moral hazard but 1% depreciation after default.

- Agents borrow to maximum with no default cost, but not necessarily so with default costs.

Welfare

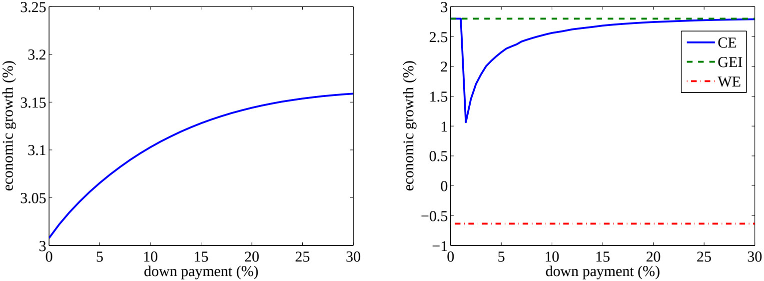


(e) $\alpha = 0$ (no moral hazard)

(f) $\alpha = 0.1$ (moral hazard)

- Individual welfare monotonically improves and approaches complete markets with no moral hazard.
- Not necessarily so with moral hazard.

Growth

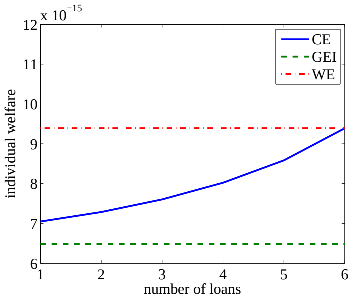


(g) $\alpha = 0$ (no moral hazard)

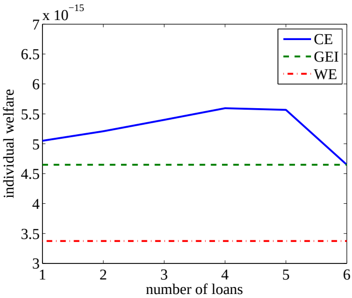
(h) $\alpha = 0.1$ (moral hazard)

- Economic growth decelerates with lower down payment.
- Intuition: risk sharing $\uparrow \implies$ precautionary saving $\downarrow$.

Adding new loans (low to high leverage)



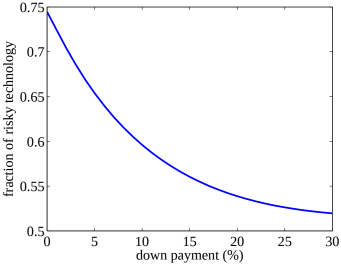
(i) $\alpha = 0$ (no moral hazard)



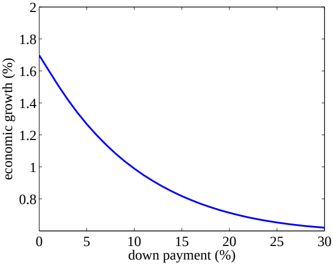
(j) $\alpha = 0.1$ (moral hazard)

- High leverage loans crowd out low ones.
- With moral hazard, may be disastrous to welfare.

Two technologies ($r_f = 0.01$, no moral hazard)



(k) Fraction of investment in the risky technology.



(l) Economic growth.

- Unlike the case with 1 technology, growth ↑ because agents allocate more capital to the risky but high return technology.

Conclusion

1. Can ex ante identical agents share idiosyncratic risk without requiring a lot of information?
   - No, if agents are not allowed to default (GEI).
   - Yes, if agents can default (Collateral equilibrium).

   Intuition: default and securitization create useful assets in nonzero net supply.
2. Moderate level of financial intermediation (leverage) is good for welfare and (typically) growth.
3. Too much leverage is bad for welfare if moral hazard issue, but competition among intermediaries endogenously lead to high leverage (subprime loans).

Relevance of investment (multiplicative) shocks

- Bertaut & Starr-McCluer (2002) find:
    - 28.2% of household wealth is primary residence,
    - 27% of household wealth is private equity.

  $\implies$ more than 50% of household wealth subject to idiosyncratic risk.
- 75% of all private equity is owned by households for which it constitutes at least 50% of their total net worth (Moskowitz & Vissing-Jorgensen, 2002).
- Consumption inequality (variance of log consumption) increases linearly with age (Deaton & Paxson, 1994).
  $\implies$ hard to explain by transitory shocks.

Collateral requirement cannot be too loose

Proposition

If $A_i^j > 0$ with positive probability for some j and $\sum_j c^{jl} < 1$ for some l, then the optimal portfolio problem has no solution.

Proof.

If $\sum_j c^{jl} < 1$, for any portfolio $(\theta, \phi, \psi) \in \Pi$ and $\epsilon > 0$,
borrow ϵ more from loan l, put up collateral $\epsilon \mathbf{c}^l$,
and invest the rest $(\epsilon(1 - \sum_j c^{jl}) > 0)$in investment j.
Then utility increases. $\square$

$\sum_j c^{jl} \geq 1$ is a natural margin constraint (leverage $\leq \infty$)
or a natural down payment constraint (down payment ≥ 0).